Sampling-period design windows for digital PI/PID control under dead-time uncertainty

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Abstract

Digital PI/PID sampling periods are commonly selected from bandwidth or rise-time rules, which mainly guard against slow sampling. This paper instead formulates sampling-period selection as a two-sided feasibility problem for simple internal model control (SIMC)-tuned first-order-plus-time-delay (FOPTD) and second-order-plus-time-delay (SOPTD) loops. Phase-margin, load-disturbance and mode-resolution requirements define upper bounds; model-coefficient resolution, derivative-noise and optional sampling-zero-cancellation requirements define lower bounds. Hold, computation, communication and filter lags are combined in an effective-delay budget. For FOPTD plants, an exact pulse-transfer model is derived for arbitrary dead time, and backward-Euler and Tustin PI implementations are compared with the half-sample-delay surrogate. A strictly convex surrogate cost has a unique unconstrained minimiser, and its feasible solution is the projection onto the sampling window. For a fixed controller designed at an estimated dead time, sampling-induced margin erosion is deterministic; dead-time uncertainty enters an absolute-phase-margin chance constraint through the upper uncertainty quantile. Across 48 plant, tuning, implementation and discretisation cases, the analytical phase-loss bound had a worst margin shortfall of less than 0.80°. In a synthetic residual-bootstrap study, the quantile design achieved 95.0% satisfaction in the surrogate and 95.2% with the exact Tustin model; backward Euler required a small numerical guard band. The method is analytical and simulation-based; controller-level fixed-point and hardware validation remain necessary.

Keywords: sampled-data control, sampling-period selection, digital PI/PID control, dead-time uncertainty, sampling zeros, coefficient quantisation

1. Introduction

The sampling period T is a small design choice with a large practical effect. In many digital PID loops it is selected after the continuous-time controller has already been tuned. A common rule is to use roughly ten to thirty samples across the closed-loop rise time, or to place the sampling frequency about one order of magnitude above the loop crossover frequency [1–3]. These rules are useful because they express the main slow-sampling effect: the zero-order hold adds phase lag and reduces the stability margin.

The same rules are also incomplete. They mainly answer the question "how slow is too slow?" They do not identify the point beyond which faster sampling provides no net benefit. That second question matters in embedded and industrial implementations. A very small sampling period can make fixed-point coefficients poorly conditioned because discrete poles cluster near $z = 1$. A discrete derivative term can amplify measurement noise. A valve or drive may be asked to update more often than is useful. For relative-degree-two plants, sampling zeros move towards $z = -1$; a controller that cancels such a zero can produce high-frequency actuator reversals. These effects are not seen in a bandwidth rule.

This paper therefore treats the sampling period as a two-sided design variable. For the common FOPTD and SOPTD process models, it forms a feasible interval $[T_{\min}, T_{\max}]$. The upper endpoint collects robustness and performance limits. The lower endpoint collects implementation limits. A strictly convex surrogate cost then gives an unconstrained candidate T_{cost}, and the final selection T_{sel} is its projection onto the feasible interval. This is not claimed to be a universal optimum. It is a structured selection rule for a specified setting: SIMC PI/PID tuning, continuous-time design followed by digital emulation, and optional retuning on the effective delay.

The contribution is deliberately narrow. First, robustness and performance requirements are expressed as upper bounds, while selected implementation limitations are expressed as lower bounds. Their intersection gives a feasible sampling-period interval rather than only a maximum allowable period. Second, the arbitrary-delay FOPTD plant is written as an exact pulse-transfer model, which permits a direct assessment of the half-sample-delay approximation and of controller discretisation. Third, a projection result gives the unique feasible minimiser of a strictly convex surrogate cost. Fourth, dead-time uncertainty is treated for the practically relevant case of a fixed controller: the chance constraint is imposed on absolute phase margin and reduces to an upper dead-time quantile. A synthetic bootstrap study then checks the probability achieved by the analytical bound and by the exact discrete models.

The remainder of the paper is organised as follows. Section 2 positions the contribution against prior work. Section 3 defines the main terms and the sampled-data model. Sections 4 and 5 derive the window and the cost-based choice for FOPTD and

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SOPTD loops respectively. Section 6 assembles the design procedure. Section 7 develops the fixed-controller absolute-margin chance constraint and the cost-invariance property. Section 8 reports the numerical checks, and Sections 9 and 10 discuss the scope and limitations.

2. Related work and positioning

Sampling rules and sampled-data foundations.. The effect of the sampling period on digital control is classical. Textbooks derive bandwidth-based rules and the zero-order-hold phase penalty [1–5]. Sampled-data theory gives the operator treatment of intersample behaviour and hold equivalents [6–10]. Related work has also studied how optimal discrete PID parameters move with the sampling period under integral time-weighted criteria [11]. Optimisation-based sampling-rate selection has been developed for LTI and multi-loop cascade systems [12], and recent work has analysed deterministic and stochastic pole-assignment controllers across candidate sampling periods [13]. These studies are mainly numerical selection tools. The present paper instead asks for closed-form feasibility endpoints under a narrower SIMC PI/PID setting.

Stability at the extremes is also well studied. On the slow side, time-delay and hybrid methods certify maximum sampling intervals, including for aperiodic sampling [14–16]. On the fast side, finite-word-length stability loss has been characterised directly [17, 18]. This paper does not re-derive those limits. Its window is a performance and implementation window placed inside them. Section 8 reports the distance to the computed discrete stability boundary for the running example.

Model-based PID tuning.. Tuning for FOPTD and SOPTD processes is mature [19]. The SIMC rules [20], based on internal model control [21], give closed-form settings with one speed–robustness knob [22]. The performance–robustness trade-off has also been mapped in detail [23]. Model-free three-term synthesis [24] is an alternative when no parametric model is available. Here the model is assumed to be an identified FOPTD or SOPTD model. Its identification uncertainty enters the design in Section 7. SIMC is used because its one-parameter form keeps the sampling analysis explicit.

Sampling zeros.. The migration of sampling zeros as $T \to 0$ is a well-understood property of sampled-data models [25, 26]: for a continuous-time plant of relative degree two, the zero-order-hold equivalent acquires a sampling zero that approaches $z = -1$ from inside the unit circle. We use this not as a general obstacle to fast sampling, but as a specific lower-bound constraint for control laws that cancel the zero.

Timing, computation delay and co-design.. Real-time control research treats delays from computation, scheduling and communication, often as random quantities. Random delays and jitter have been analysed and simulated [27–29]. Sampling and scheduling have also been posed as a co-design problem [30]. The treatment here is simpler and static. Computation and communication delays enter the effective-delay budget Eq. (3) as a constant term and a scan-proportional term. This is a form that can be evaluated at design time without a scheduling model.

Event-triggered and adaptive sampling.. A separate line abandons periodic sampling, acting only when the state crosses a threshold [31–33]. This can reduce actuation while preserving stability, but it changes both the implementation and the analysis. The present paper stays within periodic sampling, which remains the norm for industrial PLC and DCS loops, and asks how to choose the period; event-triggered schemes are an orthogonal alternative rather than a competitor.

Model-coefficient resolution.. Coefficient quantisation degrades digital controllers as the poles cluster near $z = 1$, and the delta operator was introduced to improve the conditioning at fast sampling [34]. We quantify this degradation as an explicit lower bound on T when a near-unity discrete pole or equivalent model coefficient is stored. This is not a complete finite-word-length analysis of a particular PID realisation.

Probabilistic control and identification.. Randomised and chance-constrained methods for uncertain control are well developed. Scenario methods give feasibility guarantees [35], and randomised algorithms provide a broader framework [36, 37]. The plant parameters come from system identification and may have parameter uncertainty [38]. In much of this work the sampling period is fixed before the design starts. It is not the design variable.

The closest sampling-selection work [12, 13] optimises or evaluates the sampling period numerically using performance indices. Here the output is a closed-form design window for a restricted PI/PID design class. For the upper bound, the uncertain-model result is written through the dead-time uncertainty distribution. A recent impulsive-system analysis [39] determines the maximum admissible sampling period of second-order digital PID loops, including under polynomial uncertainty. That work gives numerical and experimental support for an upper stability limit. The present paper is narrower in evidence, but it adds selected implementation lower bounds, a user-weighted interior candidate and a scalar probabilistic treatment of dead-time uncertainty. Table 1 summarises the difference.

3. Problem formulation and digital implementation

3.1. Terms used in the design window

The sampling period is denoted by T, and the sampling frequency is $f_s = 1/T$. The loop crossover frequency is ω_c, the frequency at which the nominal open-loop magnitude is one. A bandwidth or crossover rule mainly places an upper bound on T, because large T adds phase lag and can reduce robustness.

The design window in this paper is the interval $[T_{\min}, T_{\max}]$. The lower endpoint $T_{\min}$ is the largest of the fast-sampling limits, such as coefficient quantisation and derivative-noise amplification. The upper endpoint $T_{\max}$ is the smallest of the slow-sampling limits, such as phase-margin loss and load-disturbance degradation. If $T_{\min} > T_{\max}$, the chosen implementation and performance budgets are incompatible. If the window is non-empty, the unconstrained surrogate-cost minimiser T_{cost} is computed and projected onto the interval to give the feasible selection T_{sel}.

The process models are intentionally standard. FOPTD means first-order-plus-time-delay, $Ke^{-\theta s}/(\tau s + 1)$. SOPTD means second-order-plus-time-delay, with natural frequency ω_n, damping ratio ζ and dead time θ. SIMC denotes simple internal model control tuning; here it gives a PI/PID controller with a single closed-loop speed parameter λ. The analysis is for periodic sampling and emulated digital control. It is not an event-triggered or scheduling co-design method.

Table 1: Representative sampling-period selection approaches. The comparison lists the principal output and only those uncertainty or implementation effects that enter each selection calculation explicitly.

Approach	Main output	Uncertainty	Implementation and validation
Bandwidth and rise-time rules [1, 3]	Heuristic upper scale for T.	None.	Indirect implementation guidance; textbook examples.
Optimisation-based selection [12]	Numerical sampling-period choice for LTI cascade loops.	Nominal model.	Computational burden enters the objective; numerical cases.
Performance-index analysis [13]	Comparison over candidate periods for digital controllers.	Stochastic-controller setting.	Sampling-rate penalty; benchmark simulations.
Admissible-period and MSI methods [15, 39]	Stability-certified maximum period.	Selected parametric or sampling uncertainty models.	Primarily an upper-bound calculation; numerical and selected experimental cases.
Present design window	Lower and upper endpoints and projected cost choice for SIMC PI/PID loops.	Scalar dead-time chance constraint for a fixed controller.	Delay, coefficient, noise and sampling-zero effects; discrete-model, hybrid and bootstrap checks.

3.2. Zero-order hold and half-sample delay

The zero-order hold has transfer function $H(s) = (1 - e^{-sT})/s$. Its frequency response factorises exactly.

Lemma 1 (Exact hold factorisation)**.** *For all ω,*

$$H(j\omega) = \frac{1 - e^{-j\omega T}}{j\omega} = T\,\frac{\sin(\omega T/2)}{\omega T/2}\,e^{-j\omega T/2}, \tag{1}$$

so the hold magnitude is $T\,|\mathrm{sinc}(\omega T/2)|$. For $|\omega T| < 2\pi$, and hence throughout the Nyquist interval $|\omega T| \le \pi$, the sinc factor is non-negative and the hold phase is exactly $-\omega T/2$. Near the origin, $|H|/T = 1 - (\omega T)^2/24 + O((\omega T)^4)$.

The phase term in Eq. (1) is exact over the control-relevant baseband. It is not a small-ω approximation there. In phase, the hold behaves as a pure dead time of $T/2$. The magnitude droop is below 0.1 dB for $\omega T \le 0.5$. The approximation enters when this hold result is used as a surrogate for the whole sampled-data loop. That surrogate neglects hold magnitude droop, the controller discretisation, aliasing, fractional-delay details and intersample behaviour. We therefore use the working model

$$L_T(j\omega) \approx L(j\omega)\,e^{-j\omega T/2}, \tag{2}$$

in which the sampled loop is treated as the continuous loop with an added dead time of $T/2$. Figure 1 illustrates Lemma 1. The loop-level error of Eq. (2) is checked in Section 8.

3.3. Computation, communication and filtering delays

Equation (2) accounts only for the hold. A real digital loop has other lag sources. These include computation delay between sampling and actuation, communication latency, and the group delay of any analogue anti-aliasing filter. The computation delay can contain a fixed part and a period-proportional part, so we write $T_{\mathrm{comp}} = T_{\mathrm{comp},0} + c_{\mathrm{comp}}T$ with $c_{\mathrm{comp}} \in [0, 1]$. An emulation controller that applies the command within the same period has $c_{\mathrm{comp}} \approx 0$. A synchronous-scan programmable controller that applies the command at the next scan has $c_{\mathrm{comp}} \approx 1$. Collecting the constant transport terms as $T_{c0} = T_{\mathrm{comp},0} + T_{\mathrm{comm}} + T_{\mathrm{filt},0}$, the loop sees an effective dead time

$$\theta_{\mathrm{eff}} = \theta + \kappa T + T_{c0}, \qquad \kappa = \tfrac{1}{2} + c_{\mathrm{comp}} + c_{\mathrm{filt}}, \tag{3}$$

Here κ is the period-proportional coefficient, and $\Delta = \kappa T + T_{c0}$ is the added phase-lag-equivalent delay. The value $\kappa = \frac{1}{2}$ gives the ideal emulation case. The value $\kappa = \frac{3}{2}$ gives the next-scan

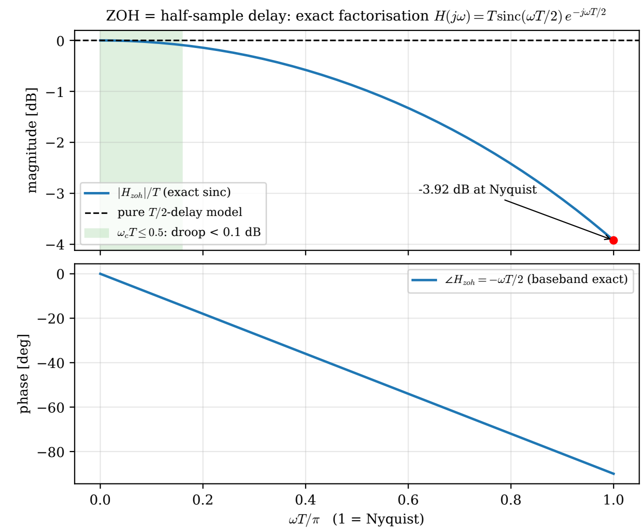

Figure 1: Zero-order hold as an effective half-sample delay (Lemma 1). The baseband phase is exactly $-\omega T/2$; the magnitude droop stays below 0.1 dB for $\omega T \leq 0.5$ and reaches -3.9 dB only at the Nyquist frequency.

case when the filter contribution is fixed. A fixed analogue anti-aliasing filter belongs in T_{c0} through its low-frequency group delay. If the filter corner is instead tied to the Nyquist frequency, for example $\omega_f = \pi/(\nu T)$, then its low-frequency delay is approximately $\nu T/\pi$ and should be included through $c_{\text{filt}} \approx \nu/\pi$. The derivations keep κ and T_{c0} explicit. A bound based on the hold alone ($\kappa = \frac{1}{2}$, $T_{c0} = 0$) is optimistic when computation delay or filter delay is present.

3.4. *Exact FOPTD pulse-transfer model and digital PI realisations*

The analytical bounds use the effective-delay surrogate, but the frequency-domain checks use an exact sampled FOPTD model. Write the process dead time as

$$\theta = mT + \delta, \qquad m = \left\lfloor \frac{\theta}{T} \right\rfloor, \qquad 0 \leq \delta < T. \tag{4}$$

For the state equation $\dot{x} = -x/\tau + (K/\tau)u(t-\theta)$, a zero-order-held input and output samples $y_k = x(kT)$ give the exact update

$$x_{k+1} = ax_k + b_0 u_{k-m} + b_1 u_{k-m-1}, \tag{5}$$

where

$$\begin{aligned} a &= e^{-T/\tau}, \\ b_0 &= K\left(1 - e^{-(T-\delta)/\tau}\right), \\ b_1 &= K\left(e^{-(T-\delta)/\tau} - e^{-T/\tau}\right). \end{aligned} \tag{6}$$

Hence the exact pulse-transfer model at the sampling instants is

$$G_T(z) = z^{-(m+1)} \frac{b_0 + b_1 z^{-1}}{1 - az^{-1}}. \tag{7}$$

Equation (7) is exact for the FOPTD process under the stated hold and sampling convention; it is not a modified-z approximation.

The PI controller is fixed in continuous time and then emulated. Backward-Euler integration gives

$$C_{\text{BE}}(z) = K_c \frac{(1 + T/\tau_I) - z^{-1}}{1 - z^{-1}}, \tag{8}$$

whereas the bilinear or Tustin rule gives

$$C_{\text{TU}}(z) = K_c \frac{(1 + r) + (-1 + r)z^{-1}}{1 - z^{-1}}, \qquad r = \frac{T}{2\tau_I}. \tag{9}$$

A next-scan implementation introduces one additional factor z^{-1}. Phase margins reported below are obtained from the first downward unit-gain crossing of $G_T(e^{j\Omega})C(e^{j\Omega})$, with continuous frequency $\omega = \Omega/T$. The derivative-noise study uses a backward difference on the measured output; the filtered case uses the explicitly stated first-order derivative filter. These definitions make the numerical checks reproducible and separate the hold effect from controller-discretisation error.

4. FOPTD loops: bounds and cost-based choice

Consider the first-order-plus-time-delay process

$$G(s) = \frac{K\, e^{-\theta s}}{\tau s + 1}. \tag{10}$$

4.1. *Nominal SIMC loop*

The SIMC rules [20] set a PI controller $C(s) = K_c\,(1 + 1/(\tau_I s))$ with

$$K_c = \frac{\tau}{K(\lambda + \theta)}, \qquad \tau_I = \min\{\tau,\ 4(\lambda + \theta)\}, \tag{11}$$

where λ (equivalently τ_c) is the closed-loop time-constant knob. When $\tau_I = \tau$ the loop transfer function reduces to a first-order plus delay integrator,

$$L(s) = \frac{e^{-\theta s}}{(\lambda + \theta)s}, \qquad \omega_c = \frac{1}{\lambda + \theta}, \qquad \text{PM}_0 = \frac{\pi}{2} - \frac{\theta}{\lambda + \theta}. \tag{12}$$

The common robust choice $\lambda = \theta$ gives $\omega_c = 1/2\theta$, $\text{PM}_0 = 61.4°$ and a gain margin of π (10 dB). The running example uses $K = 1$, $\tau = 10$ s, $\theta = 2$ s and $\lambda = \theta$. Then $\tau_I = \tau$ (since $4(\lambda + \theta) = 16\,\text{s} > \tau$) and the loop follows Eq. (12).

4.2. *Phase-margin upper bound*

The first upper bound is a robustness bound. Sampling, computation and filtering delay add phase lag near crossover. If the allowed loss of phase margin is fixed in advance, this immediately limits T. Inserting the effective delay into the loop contributes $-\omega_c \Delta$ radians at crossover.

Proposition 1 (Margin erosion). *For the SIMC loop Eq.* (12)*, the delay-equivalent surrogate Eq.* (2) *gives*

$$\text{PM}(T) = \text{PM}_0 - \omega_c \Delta = \text{PM}_0 - \frac{\kappa T + T_{c0}}{\lambda + \theta}. \tag{13}$$

Allowing a margin loss of at most $\Delta\varphi_a$ gives the upper bound

$$\kappa T + T_{c0} \ \leq\ \Delta\varphi_a(\lambda + \theta), \tag{14}$$

Table 2: Margin-erosion upper bound for the FOPTD loop Eq. (12) with $\lambda = \theta$, ideal case. Here $\omega_s/\omega_c = 2\pi/(\omega_c T)$ is the sampling-to-crossover ratio.

$\Delta\varphi_a$	$\omega_c T$	ω_s/ω_c	T/θ
5°	0.175	36	0.35
10°	0.349	18	0.70
15°	0.524	12	1.05

hence

$$T_{\max}^{\mathrm{PM}} = \frac{\Delta\varphi_a(\lambda+\theta) - T_{c0}}{\kappa}, \tag{15}$$

which reduces to $2\Delta\varphi_a(\lambda+\theta)$ in the ideal case $\kappa = \frac{1}{2}$, $T_{c0} = 0$, and contracts to $\frac{2}{3}\Delta\varphi_a(\lambda+\theta)$ for a next-scan controller ($\kappa = \frac{3}{2}$). The coefficient $1/\kappa$ multiplies the entire bracket, a point that recurs in Proposition 6.

Within the surrogate, the margin reduction in Eq. (13) is exact because a pure delay changes phase without changing crossover magnitude. The modelling error comes from replacing the complete sampled-data loop by Eq. (2), not from linearising the delay phase.

Table 2 evaluates Eq. (14) in the ideal case. It expresses the classical range $\omega_c T \approx 0.15$–0.5 (equivalently $\omega_s \approx 12$–$36\,\omega_c$) as a phase budget of about 4°–15°. It also shows how a synchronous scan tightens the bound. An analogue prefilter can do the same. If the filter is fixed, its low-frequency group delay contributes to T_{c0}. If its corner tracks the Nyquist frequency, for example ω_N/ν, it contributes approximately $c_{\text{filt}}T$ with $c_{\text{filt}} \approx \nu/\pi$. This is of the same order as the hold term and should be included in κ.

4.3. Load-disturbance upper bound

For a unit step load disturbance entering additively at the plant input, with the error convention $e = r - y$, a loop with integral action has the integrated error

$$IE = \int_0^\infty e(t)\,dt = \frac{\tau_I}{K_c}, \tag{16}$$

up to the sign set by the disturbance direction. We use its magnitude. If $e(t)$ retains one sign after the disturbance, then $IAE = |IE|$. If the error changes sign, then $IAE > |IE|$ and the formulas below are lower summaries rather than exact IAE values. If the digital controller is retuned on the effective delay, every occurrence of the identified dead time in the SIMC rule is replaced by θ_{eff}:

$$K_c = \frac{\tau}{K(\lambda+\theta_{\text{eff}})}, \qquad \tau_I = \min\{\tau, 4(\lambda+\theta_{\text{eff}})\}.$$

For the common robust SIMC choice, the tuning knob is also set to $\lambda = \theta_{\text{eff}}$. The dependence on θ_{eff} then has two useful closed forms. To avoid a circular branch definition, the branch is selected at the no-sampling baseline $\theta_{\text{eff}} = \theta_0$; if a proposed period crosses the branch boundary, the piecewise expression should be evaluated directly.

Proposition 2 (Two-regime IAE degradation)**.** *With the retuned SIMC rule above and $\lambda = \theta_{\text{eff}}$,*

- *balanced or delay-dominant at the no-sampling baseline ($\tau \le 4(\lambda+\theta_0)$, so $\tau_I = \tau$): $IE = 2K\theta_{\text{eff}}$, linear in θ_{eff};*
- *lag-dominant at the no-sampling baseline ($\tau > 4(\lambda+\theta_0)$, so $\tau_I = 4(\lambda+\theta_{\text{eff}})$): $IE = 16K\theta_{\text{eff}}^2/\tau$, quadratic in θ_{eff}.*

Let $\theta_0 = \theta + T_{c0}$ denote the no-sampling baseline when the fixed implementation delay is unavoidable. Budgeting a relative degradation ε against this baseline, that is $\theta_{\text{eff}} \le (1+\varepsilon)\theta_0$ in the linear regime and $\theta_{\text{eff}}^2 \le (1+\varepsilon)\theta_0^2$ in the quadratic regime, gives

$$\begin{aligned} T_{\max}^{IAE} &= \frac{\varepsilon\theta_0}{\kappa} && \textit{(linear)},\\ T_{\max}^{IAE} &= \frac{\theta_0(\sqrt{1+\varepsilon}-1)}{\kappa} && \textit{(quadratic)}. \end{aligned} \tag{17}$$

which reduce in the ideal case $\kappa = \frac{1}{2}$, $T_{c0} = 0$ to $2\varepsilon\theta$ and $2\theta(\sqrt{1+\varepsilon}-1) \approx \varepsilon\theta$ respectively.

For the balanced running example ($\theta = 2$ s), a 25 % IAE budget permits $T \le 1.0$ s. For a lag-dominant process ($\theta = 1$ s, $\tau = 20$ s), the quadratic law tightens this to $T \le 0.24$ s. In lag-dominant loops, this performance bound can be more restrictive than the margin bound.

4.4. Fast-sampling lower bounds

In exact arithmetic, the emulated loop converges to its continuous counterpart as $T \to 0$ [1, 6]. Thus there is no stability-based lower bound on the sampling period in the ideal model. With finite precision, however, stability itself can eventually be lost [17, 18]. The bounds below are not meant to sit at that limit. They are accuracy and noise budgets that should bind first. Each lower endpoint is therefore an implementation effect, not a stability limit.

Model-coefficient resolution.. The first lower bound comes from storing a near-unity discrete pole with finite precision. The discrete lag pole is $p = e^{-T/\tau}$, and the lag inferred from it is $\hat{\tau} = -T/\ln p$. The sensitivity of the reconstructed lag to the stored pole is

$$S = \left|\frac{\partial \ln\hat{\tau}}{\partial \ln p}\right| = \frac{1}{|\ln p|} = \frac{\tau}{T}, \tag{18}$$

This sensitivity diverges as $T \to 0$ because the pole approaches $z = 1$. A B-fractional-bit fixed-point coefficient has half-quantisation error $|\delta p| \approx 2^{-(B+1)}$. The corresponding relative model-coefficient error is $(\tau/T)2^{-(B+1)}$. Requiring this error to stay below ε_c gives

$$T_{\min}^{\text{pole}} = \frac{\tau\, 2^{-(B+1)}}{\varepsilon_c}. \tag{19}$$

For $\tau = 10$ s, $B = 12$ and $\varepsilon_c = 2$ %, this is 61 ms. For $B = 16$, it falls to 3.8 ms. The mechanism is pole clustering near $z = 1$. Increasing the number of fractional bits or using the delta operator reduces the problem. This is not a general finite-word-length stability bound for a digital PI/PID controller. Such a bound would require the chosen controller realisation, scaling, accumulator precision, rounding rules and closed-loop coefficient sensitivities.

Derivative noise.. The second lower bound applies when the controller contains derivative action. An unfiltered discrete derivative acting on white measurement noise of variance σ_n^2 produces control variance

$$\mathrm{Var}(u_D) = 2\left(\frac{K_c \tau_D \sigma_n}{T}\right)^2 \propto T^{-2}, \tag{20}$$

A derivative filter with time constant τ_D/N caps this growth at small T. Imposing an actuator-variance budget $\mathrm{Var}(u_D) \le V_{u,\max}$ turns Eq. (20) into

$$T_{\min}^{D} = \frac{\sqrt{2}\, K_c \tau_D \sigma_n}{\sqrt{V_{u,\max}}}. \tag{21}$$

Here $V_{u,\max}$ is a variance budget. If the actuator limit is specified as an RMS command contribution, its square should be used in Eq. (21). For a PI controller this term is absent. For PID it is a lower-bound constraint. Together with Eq. (19), it contributes to the lower endpoint used in Algorithm 1.

Actuator cost.. The third lower-bound mechanism is actuator activity. The number of control updates over a fixed horizon scales as $1/T$. For a control law that cancels a sampling zero near $z = -1$, the total variation of the command also scales as $1/T$ (Section 5). Either way, faster sampling can increase actuator activity even when the output response looks acceptable.

4.5. Cost-based choice inside the window

The bounds define feasibility but do not select a point inside the interval. Let $\theta_0 = \theta + T_{c0}$ and consider the update-penalised surrogate

$$J_u(T) = \alpha(\theta_0 + \kappa T)^2 + \frac{\gamma}{T}, \qquad T > 0. \tag{22}$$

With the dimensionless variable $y = \kappa T/\theta_0$ and

$$\rho_u = \frac{\gamma\kappa}{2\alpha\theta_0^3}, \tag{23}$$

the stationarity condition is

$$y^2(1 + y) = \rho_u. \tag{24}$$

The left side is strictly increasing for $y > 0$, and $J_u''(T) = 2\alpha\kappa^2 + 2\gamma/T^3 > 0$; therefore, a unique positive minimiser T_{cost} exists. For $\rho_u \ll 1$, $y \sim \sqrt{\rho_u}$; for $\rho_u \gg 1$, $y \sim \rho_u^{1/3}$. The ideal running example has $\kappa = 1/2$, $T_{c0} = 0$ and the earlier parameter $\rho_1 = \gamma/(\alpha\theta^3) = 0.02$, which corresponds to $\rho_u = 0.005$. Equation (24) gives $T_{\mathrm{cost}} = 0.274$ s. This lies inside $[T_{\min}^{\mathrm{pole}}, T_{\max}^{IAE}] = [0.061\text{ s}, 1.0\text{ s}]$.

For a derivative-noise penalty, use

$$J_d(T) = \alpha(\theta_0 + \kappa T)^2 + \frac{\beta}{T^2}. \tag{25}$$

Defining $\rho_d = \beta\kappa^2/(\alpha\theta_0^4)$ yields

$$y^3(1 + y) = \rho_d, \tag{26}$$

which also has one positive root. These costs are user-weighted selection criteria; they are not universal performance indices or physical wear laws.

5. SOPTD loops: damping, aliasing and sampling zeros

Consider the second-order-plus-time-delay process

$$G(s) = \frac{K\omega_n^2\, e^{-\theta s}}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \tag{27}$$

The damping ratio changes the sampling problem. Overdamped plants reduce to real lag factors and largely follow the FOPTD logic. Underdamped plants add an oscillatory mode, and the sampling period must also resolve that mode.

5.1. Damped-mode frequency and aliasing

For $\zeta < 1$, the damped natural frequency is $\omega_d = \omega_n\sqrt{1-\zeta^2}$. The -3 dB bandwidth follows from $u^2 - 2(1-2\zeta^2)u - 1 = 0$ with $u = (\omega_b/\omega_n)^2$, giving

$$\omega_b = \omega_n\left[(1-2\zeta^2) + \sqrt{(1-2\zeta^2)^2 + 1}\right]^{1/2}, \tag{28}$$

This equals ω_n at $\zeta = 1/\sqrt{2}$ and $1.55\,\omega_n$ as $\zeta \to 0$. The zero-order-hold equivalent has poles $z_{1,2} = e^{-\zeta\omega_n T} e^{\pm j\omega_d T}$. The exponential map is not injective, so modes at ω_d and $\omega_d - 2\pi k/T$ coincide. If $\omega_d T = 2\pi$, the sampled step response becomes $y(kT) = 1 - e^{-\zeta\omega_n kT}$. This is monotone, while the continuous plant rings between samples. The digital loop then misses the resonance. This gives the Nyquist mode-resolution limit $\omega_d T < \pi$. For fidelity, a stricter rule such as eight to ten samples per damped period, $\omega_d T \lesssim \pi/4$, keeps the poles away from the negative real axis. Figure 2 shows the aliased case.

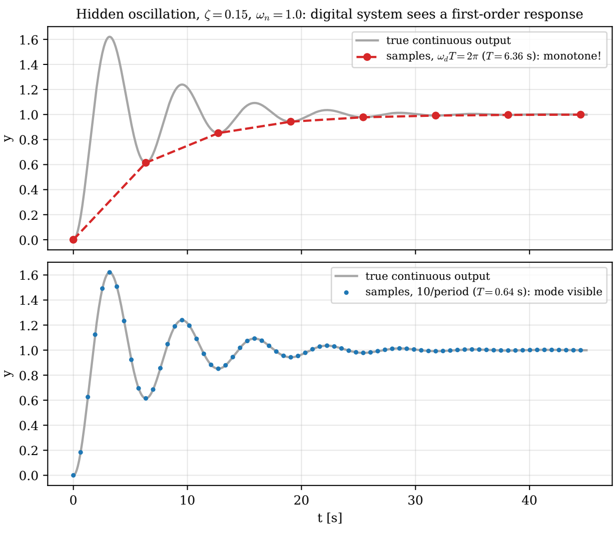

Figure 2: Hidden oscillation for an underdamped SOPTD plant ($\zeta = 0.15$, $\omega_n = 1$). Sampling at $\omega_d T = 2\pi$ (top) yields a monotone sequence that hides a strongly ringing response; ten samples per period (bottom) resolves the mode.

5.2. Overdamped plants

For $\zeta \ge 1$, the plant Eq. (27) factorises into two real lags. A series-form SIMC PID is usually assigned so that the PI zero is associated with the dominant lag and the derivative zero with the smaller, faster lag; the SIMC half rule can also reduce the plant first to an FOPTD model [20]. Under those standard reductions, the loop is treated by the FOPTD window of Section 4 with the effective delay in Eq. (3). No additional sampling-zero-cancellation constraint is introduced unless the final discrete controller deliberately cancels the sampled zeros.

5.3. Underdamped plants: sampling zeros and derivative noise

For $\zeta < 1$, the poles are complex. Direct cancellation is not generally available. An ideal-form PID can place complex zeros only if $\tau_I < 4\tau_D$. Even then, cancelling a lightly damped mode is usually a poor choice, because the mode remains excited through the load path. The design constraints are therefore the mode-fidelity condition $\omega_d T \lesssim \pi/4$, the margin bound at the chosen crossover, the derivative-noise bound Eq. (21), and the sampling-zero constraint below.

For the delay-free second-order factor in Eq. (27), and for the full plant when θ/T is an integer, the zero-order-hold discretisation (with $a = \zeta\omega_n$, $\varphi = \omega_d T$, $E = e^{-aT}$) has numerator $K(b_1 z + b_0)$ with

$$b_1 = 1 - E\Big(\cos\varphi + \tfrac{a}{\omega_d}\sin\varphi\Big), \tag{29}$$

$$b_0 = E^2 - E\Big(\cos\varphi - \tfrac{a}{\omega_d}\sin\varphi\Big). \tag{30}$$

Proposition 3 (Sampling-zero migration)**.** *The sampling zero $z_0 = -b_0/b_1$ of the delay-free relative-degree-two factor satisfies*

$$z_0 = -1 + \tfrac{2}{3}\zeta\omega_n T + O(T^2), \tag{31}$$

so $z_0 \to -1$ as $T \to 0$, on the unit circle.

Proof. Taylor expansion of the numerator coefficients gives $b_1 = \frac{1}{2}\omega_n^2T^2 - \frac{1}{3}\zeta\omega_n^3T^3 + O(T^4)$ and $b_0 = \frac{1}{2}\omega_n^2T^2 - \frac{2}{3}\zeta\omega_n^3T^3 + O(T^4)$. Substitution in $z_0 = -b_0/b_1$ gives Eq. (31). □

For a non-integer process dead time, the pulse-transfer model contains an additional fractional-delay contribution. The finite zero above remains the sampling zero of the dynamic second-order factor, but it should not be called the complete plant zero without specifying the fractional-delay model.

This creates a lower-bound constraint for zero-cancelling designs, such as deadbeat, minimum-variance or naive pole-placement designs. A controller pole placed at z_0 makes the command ring at the Nyquist frequency. The ring decays by $|z_0|$ per sample. Using Eq. (31), the small-T ringing time is asymptotically $t_{\mathrm{ring}} \sim 3\ln(1/\varepsilon)/(2\zeta\omega_n)$, which is independent of T to first order. Faster sampling therefore raises the commanded ringing frequency and increases drive activity. A practical response is to avoid cancelling z_0, or to impose an explicit tolerance such as $|z_0| \le 0.9$. The convenient rule $\zeta\omega_n T \gtrsim 0.15$ is only this selected tolerance expressed through the first-order expansion; it is practical guidance, not an independent theorem. Figure 3 compares the computed locus with Eq. (31). Figure 4 shows the resulting command variation.

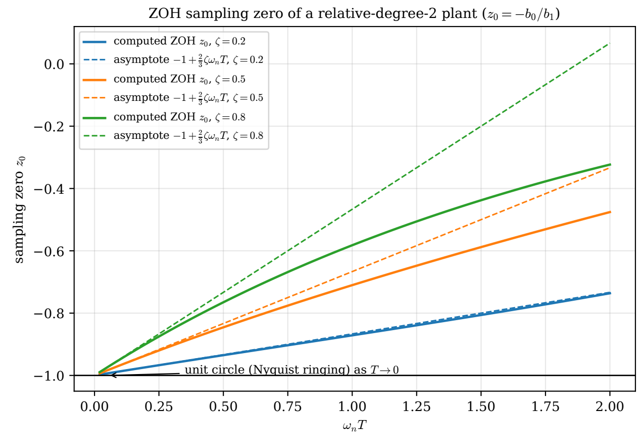

Figure 3: Sampling zero of the relative-degree-two plant against the first-order expansion Eq. (31). The zero converges to $z = -1$ as $T \to 0$ for every damping ratio.

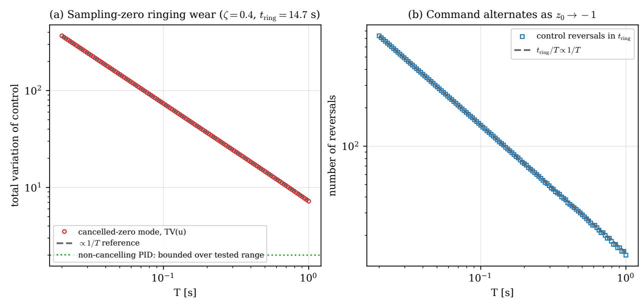

Figure 4: Command variation for a cancelled sampling-zero mode ($\zeta = 0.4$). Total variation (a) and the number of control reversals (b) both scale as $1/T$; a non-cancelling PID keeps the variation bounded and approximately T-independent over the tested range.

Algorithm 1: Constraint-based sampling-period selection

Input: Plant and controller model; delay architecture; performance, precision and noise budgets; uncertainty quantile if used; surrogate-cost weights.
Output: T_{sel}, $[T_{\min}, T_{\max}]$, T_{cost} and active constraints.
Compute all applicable positive upper bounds and record their minimum $T_{\max}$;
If a chance-bound numerator in Eq. (39) is non-positive, report infeasibility even as $T \to 0$ and stop;
Compute all applicable lower bounds and record their maximum $T_{\min}$;
if $T_{\min} > T_{\max}$ **then**
  report infeasibility and the active conflicting constraints; stop;
Solve Eq. (24) or Eq. (26) for T_{cost};
Set T_{sel} from Eq. (33);
Validate the selected period with Eq. (7) and the chosen digital controller;
return T_{sel}, $[T_{\min}, T_{\max}]$, T_{cost}, *active constraints*;

5.4. Cost-based choice with derivative noise

For PID loops the derivative-noise surrogate J_d in Eq. (25) is used. The unique root of Eq. (26) defines T_{cost}, which is then projected onto the feasible interval. For $\omega_n = 1$, $\zeta = 0.4$, $\theta = 0.5$ s and the original ideal-case ratio $\beta/(\alpha\theta^4) = 0.01$, the result is $T_{\text{cost}} = 0.13$ s. At this period the delay-free sampling zero is approximately -0.965 and should not be cancelled.

6. Feasible-window selection procedure

Collect the applicable lower and upper constraints as

$$T_{\min} = \max_i T_{\min,i}, \qquad T_{\max} = \min_j T_{\max,j}. \tag{32}$$

If $T_{\min} > T_{\max}$, the requested performance and implementation budgets are incompatible.

Proposition 4 (Projection of a strictly convex cost)**.** *Let J be strictly convex on* $(0, \infty)$ *with unique unconstrained minimiser T_{cost}, and let the feasible interval* $[T_{\min}, T_{\max}]$ *be non-empty. The unique constrained minimiser is*

$$T_{\text{sel}} = \operatorname{proj}_{[T_{\min},T_{\max}]}(T_{\text{cost}}) = \min\{T_{\max}, \max\{T_{\min}, T_{\text{cost}}\}\}. \tag{33}$$

Proof. Strict convexity makes J' strictly increasing and changes its sign once, at T_{cost}. If T_{cost} lies inside the interval it is feasible and optimal. If it lies below the interval, J is increasing throughout the feasible set and the minimum is $T_{\min}$; if it lies above, J is decreasing throughout the feasible set and the minimum is $T_{\max}$. □

7. Dead-time uncertainty and fixed-controller chance feasibility

Let $\widehat{\theta}$ denote the dead time used to tune a fixed controller, while θ denotes the uncertain plant dead time. In the regime $\tau_I = \tau$, the implemented controller has

$$K_c = \frac{\tau}{K(\lambda + \widehat{\theta})}, \tag{34}$$

so the actual continuous open loop is

$$L(s;\theta) = \frac{e^{-\theta s}}{(\lambda + \widehat{\theta})s}, \qquad \omega_c = \frac{1}{\lambda + \widehat{\theta}}. \tag{35}$$

This distinction matters: a controller cannot be retuned to every unknown realisation of θ after implementation.

Proposition 5 (Deterministic sampling-induced erosion for a fixed controller)**.** *Under the delay-equivalent model, the phase margin is*

$$\mathrm{PM}(T,\theta) = \frac{\pi}{2} - \frac{\theta + \kappa T + T_{c0}}{\lambda + \widehat{\theta}}. \tag{36}$$

Consequently,

$$\mathrm{PM}(0,\theta) - \mathrm{PM}(T,\theta) = \frac{\kappa T}{\lambda + \widehat{\theta}}, \tag{37}$$

which is independent of the uncertain dead time. Dead-time uncertainty therefore does not create a non-trivial chance constraint on sampling-induced erosion for a fixed controller.

A meaningful uncertainty requirement is instead imposed on the absolute phase margin. Let $q_{1-\alpha}(\theta)$ be the upper $(1-\alpha)$ quantile of the dead-time uncertainty distribution and define $A = \pi/2 - \mathrm{PM}_{\min}$.

Proposition 6 (Fixed-controller absolute-margin chance bound)**.** *The chance constraint*

$$\Pr\{\mathrm{PM}(T,\theta) \geq \mathrm{PM}_{\min}\} \geq 1 - \alpha \tag{38}$$

is equivalent, under Eq. (36)*, to*

$$T \leq T^{\text{cc}}_{\max}(\alpha) = \frac{A(\lambda + \widehat{\theta}) - q_{1-\alpha}(\theta) - T_{c0}}{\kappa}. \tag{39}$$

If the numerator is non-positive, the requested confidence and phase margin are infeasible even as $T \to 0$.

Proof. Equation (36) is decreasing in θ. The event in Eq. (38) is equivalent to $\theta \leq A(\lambda + \widehat{\theta}) - \kappa T - T_{c0}$. Its probability is at least $1-\alpha$ exactly when this threshold is no smaller than $q_{1-\alpha}(\theta)$. Rearranging gives Eq. (39). □

Lemma 2 (Dispersion invariance of the selected quadratic surrogate)**.** *If θ has mean $\bar{\theta}$ and variance σ^2_θ, and $J(T,\theta) = \alpha(\theta + \kappa T + T_{c0})^2 + g(T)$ with deterministic g, then*

$$\mathbb{E}[J(T,\theta)] = \alpha(\bar{\theta} + \kappa T + T_{c0})^2 + g(T) + \alpha\sigma^2_\theta. \tag{40}$$

Thus the unconstrained minimiser is independent of σ^2_θ. This property is lost if cost weights or other uncertain parameters depend on the identified model.

Table 3: Constraint summary. Actuator update or wear is a soft cost unless an explicit admissible budget is supplied.

Constraint	Direction	Source or condition
Margin erosion	upper	Proposition 1
Load-disturbance IAE	upper	Proposition 2
Mode resolution	upper	$\omega_d T < \pi$ or a stated fidelity rule
Model-coefficient resolution	lower	Eq. (19)
Derivative noise	lower	Eq. (21)
Sampling-zero cancellation	lower, optional	exact $\lvert z_0(T)\rvert$ tolerance; rules such as $\zeta\omega_n T \gtrsim 0.15$ are heuristic
Update or wear penalty	soft	Eq. (22)

The chance bound affects the upper endpoint, while Lemma 2 concerns only the location of the unconstrained surrogate-cost minimiser. The practical selection remains Eq. (33), with $T_{\max}$ replaced by the minimum of the deterministic and chance-constrained upper bounds.

8. Numerical validation and benchmark study

The accompanying script `validation_ejc.py` generates the new pulse-transfer, benchmark and bootstrap results with fixed random seed 10. The sampled FOPTD response uses Eq. (7); phase margins use the first downward gain crossover. Existing IAE, coefficient-resolution, derivative-noise and sampling-zero checks are retained as mechanism-level verification.

8.1. *Exact pulse-transfer check of the delay surrogate*

Figure 5 compares the delay-equivalent margin with the exact pulse-transfer model for the running example. Backward-Euler and Tustin PI realisations are both shown, together with immediate-update and next-scan timing. Over the conventional phase-budget range, the discrepancy is below approximately one degree for this running-example sweep. This statement is separate from the multi-case design-point benchmark below, where some backward-Euler cases are conservative by several degrees.

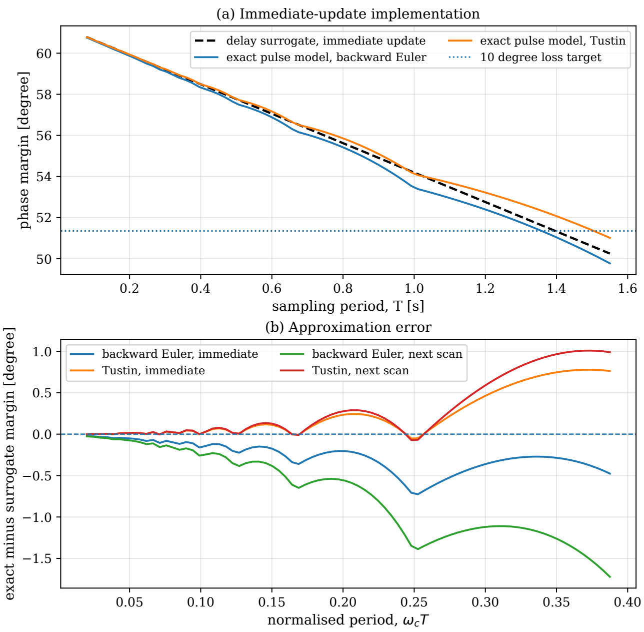

Figure 5: Exact pulse-transfer validation for the FOPTD running example. (a) Immediate-update phase margin for the delay surrogate, backward-Euler PI and Tustin PI. (b) Exact-minus-surrogate margin error for immediate-update and next-scan implementations.

8.2. *Multi-case phase-budget benchmark*

The analytical 10° phase-loss bound was evaluated over six dead-time ratios $\theta/\tau \in \{0.15, 0.20, 0.30, 0.50, 0.75, 1.00\}$, two tuning ratios $\lambda/\theta \in \{1, 2\}$, two timing architectures and two PI discretisations, giving 48 cases. All cases satisfy $\tau_I = \tau$, the domain of Proposition 1. Figure 6 and Table 4 compare the exact pulse-transfer margin with the analytical target. The largest negative error, which is the relevant margin shortfall, was 0.80°; larger positive errors indicate conservatism rather than a violated margin budget. Tustin with explicit next-scan delay followed the delay surrogate most closely in this benchmark.

8.3. *Load-disturbance, precision and noise checks*

For the lag-dominant FOPTD example, simulated IAE divided by the analytical value in Proposition 2 remains between 1.000 and 1.002 across the tested periods. Figure 7 shows the result.

The model-coefficient calculation follows the sensitivity τ/T in Eq. (18); Fig. 8 shows the resulting dependence on the number of fractional bits. This remains a representation bound, not a controller-level fixed-point stability result.

Table 4: Exact pulse-transfer error at the analytical 10° phase-loss bound over 12 plant/tuning cases per implementation. A negative shortfall means that the exact phase margin fell below the analytical target.

Implementation	Median $\lvert e_{\mathrm{PM}} \rvert$ [°]	Worst shortfall [°]	Max. excess [°]
Immediate, backward Euler	1.63	-0.48	6.00
Immediate, Tustin	0.70	-0.80	1.39
Next scan, backward Euler	0.58	-0.29	2.43
Next scan, Tustin	0.05	0.00	0.12

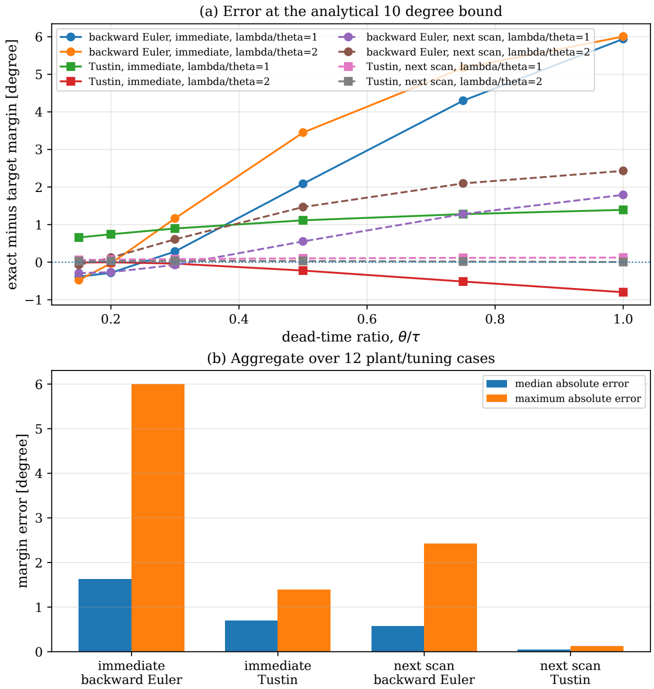

Figure 6: Phase-budget benchmark over 48 plant, tuning, timing and discretisation combinations. The ordinate is exact phase margin minus the analytical target; negative values denote shortfall and positive values conservatism.

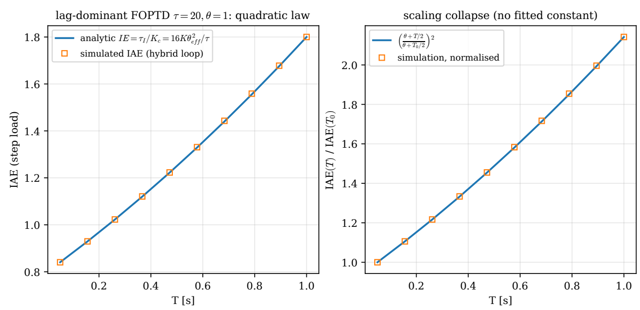

Figure 7: Load-disturbance IAE for a lag-dominant FOPTD loop ($\tau = 20$ s, $\theta = 1$ s). The hybrid simulation follows the analytical quadratic law without a fitted scale factor.

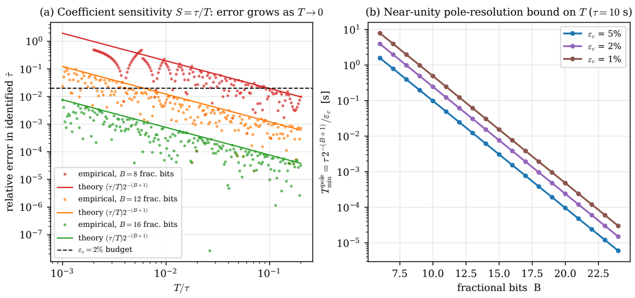

Figure 8: Model-coefficient resolution. The reconstructed-lag error follows the near-unity-pole sensitivity, and the corresponding lower period falls rapidly as the number of fractional bits increases.

The derivative-noise Monte Carlo check uses 40 replications per period. The fitted unfiltered log-slope is -1.998, close to the -2 exponent in Eq. (20); the filtered derivative has bounded high-frequency gain.

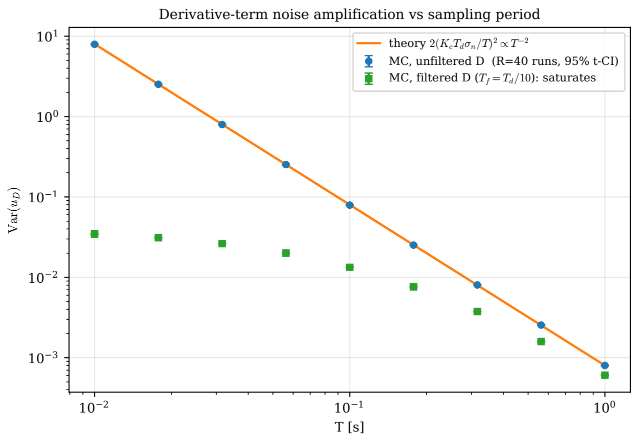

Figure 9: Derivative-term noise amplification. The unfiltered variance follows T^{-2}; a first-order filtered derivative limits the high-frequency gain.

For the sampling-zero expansion, $\zeta = 0.4$, $\omega_n = 1$ and $T = 0.35$ s give the exact delay-free zero $z_0 = -0.9106$ and the first-order value -0.9067, a 0.4% discrepancy. The command-variation implication is shown in Fig. 4; the result concerns commanded activity, not guaranteed mechanical motion.

8.4. *SOPTD sampled-data check*

For an underdamped SOPTD plant, the continuous ideal PID choice $\tau_I = 2\zeta/\omega_n$ and $\tau_D = 1/(2\zeta\omega_n)$ cancels the delay-free second-order denominator and leaves an integrator with process delay. This cancelling construction is not recommended for lightly damped plants; it is used here only to create a transparent SOPTD sampled-data check. The script splits the dead time as $\theta = mT + \delta$, forms the exact ZOH state-space model, and compares the analytical 5° phase-loss period with a Tustin digital PID over 6 underdamped cases. The maximum sampled mode angle is $\omega_d T = 0.42$, so the test is inside the mode-fidelity region. The maximum absolute phase-margin error is 0.17°, with no negative shortfall.

Table 5: SOPTD exact sampled-data check at the analytical 5° phase-loss period. The loop uses the cancellable continuous PID construction described in the text and a Tustin digital PID in the exact check.

ζ	ω_n [rad/s]	T [s]	$\omega_d T$	Error [°]
0.25	0.70	0.623	0.42	0.13
0.25	1.00	0.436	0.42	0.13
0.40	0.70	0.623	0.40	0.15
0.40	1.00	0.436	0.40	0.15
0.70	0.70	0.623	0.31	0.17
0.70	1.00	0.436	0.31	0.17

8.5. Synthetic bootstrap validation of the chance bound

A synthetic FOPTD step response with $(K, \tau, \theta) = (1, 10, 2)$, sampling interval 0.5 s and Gaussian output-noise standard deviation 0.03 was fitted by nonlinear least squares. A residual bootstrap with 1000 replications gave the point estimate $\widehat{\theta} = 2.005$ s, replicate mean $\bar{\theta}_b = 2.024$ s and upper 95% quantile $q_{0.95} = 2.251$ s. With $\lambda = \widehat{\theta}$, $T_{c0} = 0.30$ s, $\kappa = 1/2$ and $\mathrm{PM}_{\min} = 50°$, Eq. (39) gives $T^{\mathrm{cc}}_{\max} = 0.498$ s. Using the replicate mean instead of the quantile gives 0.952 s and only 50.1% satisfaction in the surrogate.

At the quantile period, the surrogate achieved 95.0% satisfaction (95% Wilson interval 93.5–96.2%). The exact pulse-transfer model gave 95.2% for Tustin (93.7–96.4%) and 92.9% for backward Euler (91.1–94.3%). The latter shortfall is discretisation error not represented by the delay surrogate; reducing the period to 0.458 s restored 95.1% empirical satisfaction (93.6–96.3%). These intervals quantify Monte Carlo uncertainty and show why the analytical chance bound should be followed by a discrete-model validation step.

8.6. Comparison with a bandwidth rule

For the running loop, a rule using N samples per crossover period gives $T_N = 2\pi/(N\omega_c)$. With $\omega_c = 0.25$ rad/s, $N = 10$ gives $T_N = 2.51$ s and a hold-only phase loss of about 18°, outside the selected 10° budget. With $N = 20$, the period is 1.26 s and the loss is about 9°, inside that budget. Thus a conventional bandwidth rule can agree with the proposed window, but only after the allowable phase-loss budget has been chosen. Maximum-sampling interval methods such as Wang et al. [39] answer a different question: they certify stability or exponential convergence for a specified sampled-data PID loop, whereas the present window is a performance and implementation selection rule inside the stability boundary.

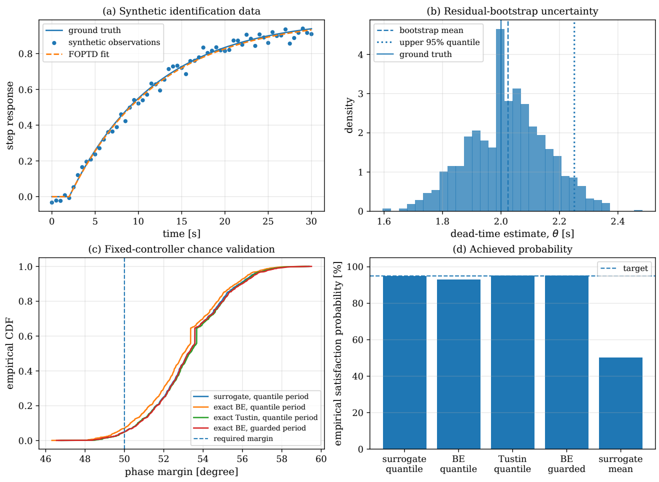

Figure 10: Synthetic uncertainty study. (a) Noisy step data and fitted FOPTD model. (b) Residual-bootstrap dead-time distribution. (c) Empirical phase-margin distributions for the quantile period and a backward-Euler guard period. (d) Achieved probability compared with the 95% target.

8.7. Synchronous-scan example

For the running loop, next-scan actuation adds one full sample delay, so the surrogate coefficient changes from $\kappa = 1/2$ to $3/2$. By the $1/\kappa$ scaling in Eq. (15), the 10° margin bound contracts from 1.40 s to 0.47 s. The update-penalised $T_{\mathrm{cost}} = 0.274$ s is the candidate computed earlier for $\kappa = 1/2$; here it is only being checked for continued feasibility, not re-optimised for the next-scan architecture. A hold-only calculation would overstate the allowable period by a factor of three.

9. Discussion and limitations

The results give a practical reading of familiar sampling rules. If the plant is well identified, the controller is PI, the controller coefficients are stored in floating point or in a sufficiently wide fixed-point word, and the actuator is not wear-limited, the lower bounds are usually slack. In that case the main question is still the upper bound, and a classical bandwidth rule is often adequate.

The same rule becomes incomplete when one of these assumptions fails. A narrow fixed-point word raises $T_{\min}^{\text{pole}}$. A PID derivative acting on a noisy measurement raises the derivative-noise lower bound. A zero-cancelling design activates the sampling-zero constraint. An uncertain dead-time estimate lowers the chance-constrained upper bound. In these cases the useful object is the whole interval $[T_{\min}, T_{\max}]$, not only $T_{\max}$. If the interval is empty, the conclusion is not merely numerical. It means that the proposed implementation and the requested performance cannot both be met. The remedy must then be physical or algorithmic: for example, more fractional bits, a stronger derivative filter, a non-cancelling design, a slower performance target or more identification data.

Three points follow. First, the hold is not always the main sampled-data lag. On a synchronous-scan controller, computation delay can equal the hold term. An anti-aliasing filter can add more delay. A hold-only bound is then optimistic. Second, faster sampling is not always harmless. Proposition 3 and Fig. 4 show this for relative-degree-two plants under cancelling designs. Third, the upper endpoint is a margin bound, not a stability certificate. In the running example the immediate-update 10° margin bound is 1.40 s, while the delay-surrogate zero-margin value is $[\frac{\pi}{2}(\lambda + \theta) - \theta]/\kappa = 8.57$ s, about 6.1 times larger. Exact discrete stability certification is still a separate problem handled by maximum-sampling-interval methods [15, 39].

The uncertainty distinction is essential. For a fixed controller, uncertain dead time changes the baseline phase margin but not the sampling-induced erosion. The meaningful chance constraint is therefore imposed on absolute phase margin and uses an upper dead-time quantile. A lower-quantile erosion bound would correspond to a family of controllers retuned for every parameter realisation, not to one implemented controller. The bootstrap example also shows that the analytical bound must be checked with the actual controller discretisation.

The principal limitations are as follows. The reduction to a single loop transfer function relies on SIMC tuning. For SOPTD plants it also relies on either overdamped factorisation or an explicitly non-cancelling underdamped design. Actuator saturation and anti-windup [40] are not modelled. The IAE law assumes a near-monotone response. The uncertainty calculation is one synthetic residual-bootstrap example and is not a substitute for application-specific identification data. Most importantly, the checks are simulation-based. A hardware study on a fixed-point target is still needed, especially for the model-coefficient and actuator-variation lower bounds.

10. Conclusion

Sampling-period selection for SIMC-tuned FOPTD and SOPTD PI/PID loops has been formulated as a two-sided feasibility problem. Robustness, load-disturbance and mode-resolution requirements supply upper limits; coefficient resolution, derivative noise and optional sampling-zero-cancellation requirements supply lower limits. A strictly convex surrogate has a unique unconstrained minimiser, and the feasible selection is its projection onto the interval.

For FOPTD plants, an exact arbitrary-delay pulse-transfer model was derived and used to separate hold, timing and controller-discretisation effects. In the 48-case phase-budget benchmark, the analytical delay bound had a worst phase-margin shortfall below 0.80°, although some backward-Euler cases were conservative by several degrees. Under dead-time uncertainty, the controller must be treated as fixed. Sampling-induced erosion is then deterministic, while an absolute-margin chance constraint reduces to the upper dead-time quantile. The synthetic bootstrap study met the nominal probability under the surrogate and Tustin model; backward Euler required a small numerical guard band.

The design window is not a stability certificate, and the surrogate cost is not a universal optimum. The remaining priorities are controller-level fixed-point analysis, application-specific identification data and hardware or hardware-in-the-loop validation.

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A. Bhattacharyya: Conceptualization, Methodology, Formal analysis, Software, Validation, Visualization, Writing – original draft, Writing – review and editing. AUTHOR ACTION REQUIRED: verify every listed role.

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Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. AUTHOR ACTION REQUIRED: verify this statement and complete Elsevier’s declarations process during submission.

Data and code availability

All figures and numerical results are generated by Python scripts using NumPy, SciPy and Matplotlib. AUTHOR ACTION REQUIRED: deposit the scripts, raw outputs, configuration, software environment and random seeds in a persistent public repository, then replace this sentence with the repository citation and DOI.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT and Codex to assist with manuscript restructuring, language editing, LaTeX conversion, literature organisation, numerical-validation support and code review. The author reviewed and verified all generated material, corrected the manuscript and software as necessary, and takes full responsibility for the content of the article.

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